4.8: Antiderivative

- Indefinite integrals involving powers and basic trig formulas. **HW 1-8**
- Initial value problems. **HW 9-15**

5.1: Approximating Areas Under the Curves

- Calculating left and right Riemann sums. **HW 2, 5-7, 9**
- Reading and writing sigma notation. **HW 10-11**

5.2: Definite Integrals.

- Evaluating definite integrals using geometry. **HW 7-9, 17**
- Properties of definite integrals
  - Know the difference between finding the *area* and *net area* of a region. **HW 1, 19**
  - Evaluating definite integrals from graphs. **HW 10-12**
  - Evaluating definite integrals from given values. **HW 13-15**

5.3: Fundamental Theorem of Calculus

- Applying the FTC, and understanding the inverse relationship between differentiation and integration. **HW 1-5, 17-18**
- Evaluating definite integrals for explicit values. **HW 7-12, 14, 19**
- Find the area bounded by a curve. **HW 13, 20**
  - Find the *net area* of the region bounded by the curve and the x-axis on an interval, i.e., the definite integral on that interval.
  - Find the *area* of the regions bounded by the curve and the x-axis on an interval, i.e., calculate all area as positive.

5.4: Working with Integrals

- Using properties of even and odd functions. **HW 1-2, 4-8, 19**
- Finding the average value of a function. **HW 9-14, 18**

5.5: Substitution Rule

- Using the substitution rule to find the indefinite integral. You *will not necessarily be told which problems require this rule.* **HW 1-10**

Some general notes.

- You will be given the necessary formulas from section 4.8.
- The quiz and homework are good reviews. You can review pass due HW assignments in MyMathLab (MML) by going to “Gradebook” and clicking on “Review”. Here you can practice more problems by clicking on “Similar Exercise”.
- More practice problems can be found in MML in “Study Plan” or your textbook.
- There will be 15-20 problems including some “easy” problems involving properties of integrals which can worked quickly.